

① Single RNN

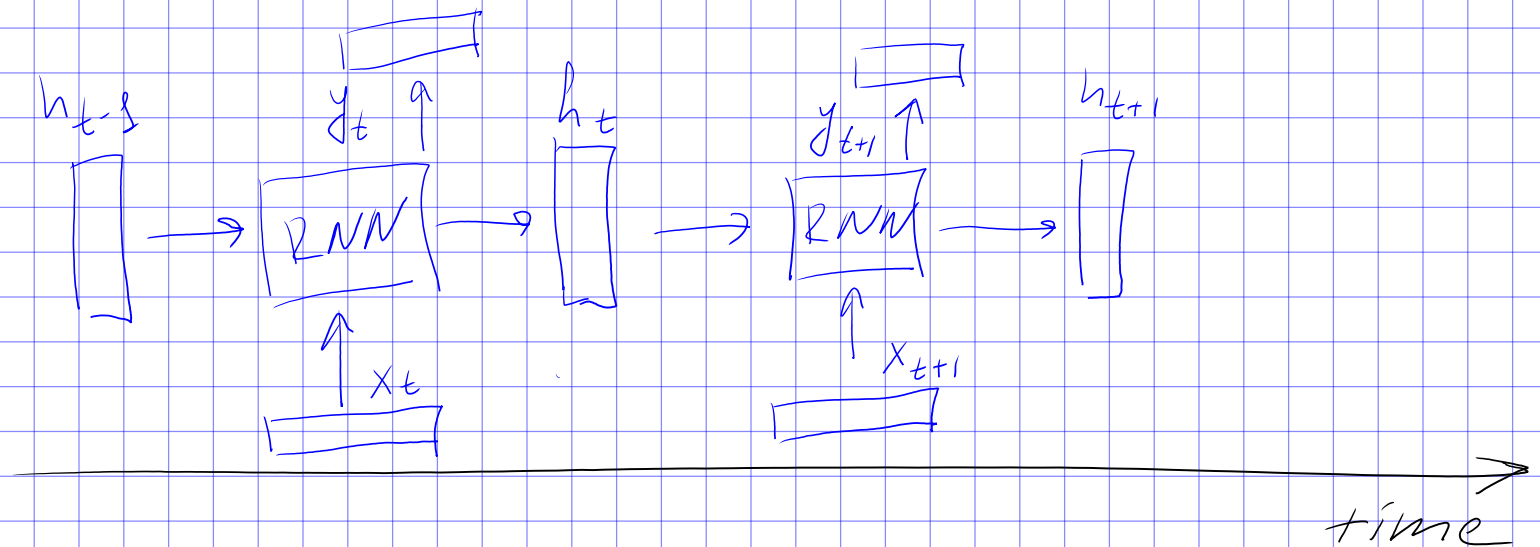
Dynamics:

a) forward: $(h_{t-1}, x_t) \xrightarrow{f} h_t$

b) output projection: $(h_{t-1}, x_t) \xrightarrow{g} y_t$

... or $h_t \rightarrow y_t$

single axis: time



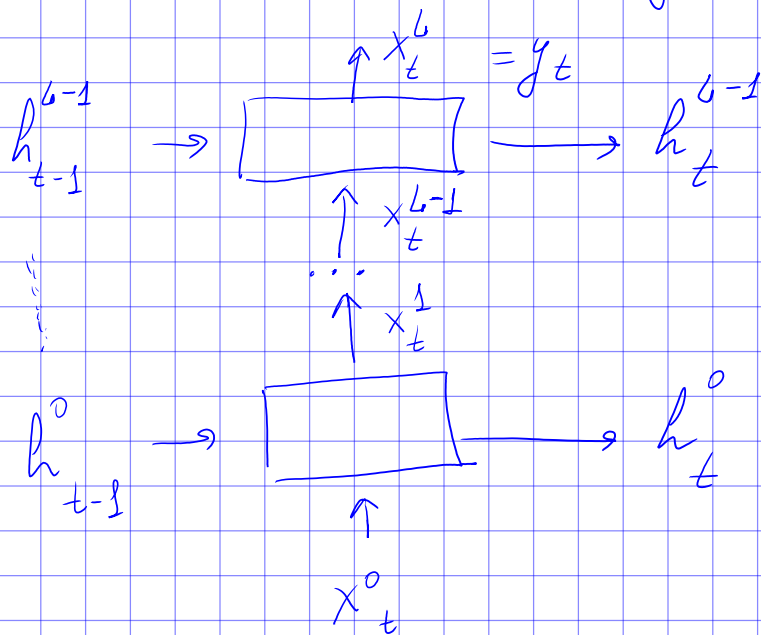
② Let's add more layers $\Rightarrow$ L-layer RNN

+ 2nd axis : hierarchy (layers)

layers $\uparrow$
time $\rightarrow$

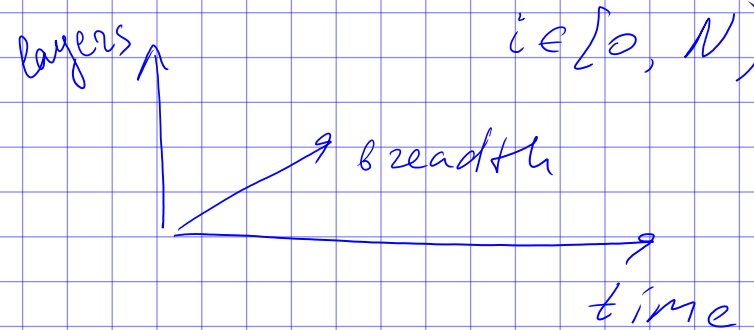
output projection $g(h_{t-1}^l, x_t^l) = y_t^l =: x_t^{l+1}$

is now
upward hierarchy
dynamics



$$H_t = \text{stack}(h_t^0, h_t^1, \dots, h_t^{L-1})$$

③ Now consider N L -layer RNNs that run
separately in parallel $\Rightarrow$ +3rd axis: breadth
 $i \in [0, N)$



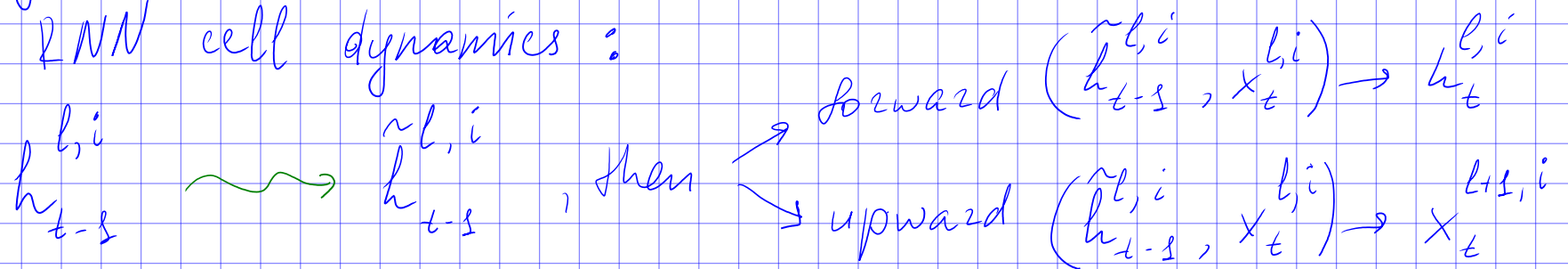
Note that each RNN cell
dynamics is still the same
as in ①

$$H_t = \text{stack2d} \left(\{ h_t^{i,l} \} \right) \\ i \in [0, N), l \in [0, L)$$

OK, N separate RNNs
is not fancy

④ Let's fuse them [i.e. parallel RNNs]
 using some sort of attention:
 self-attention (SA) or cross-attention (CA)

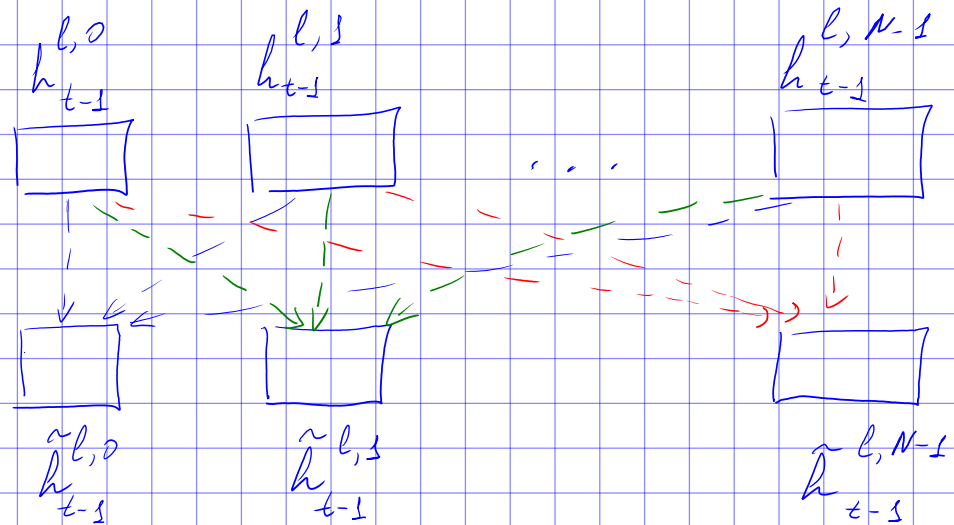
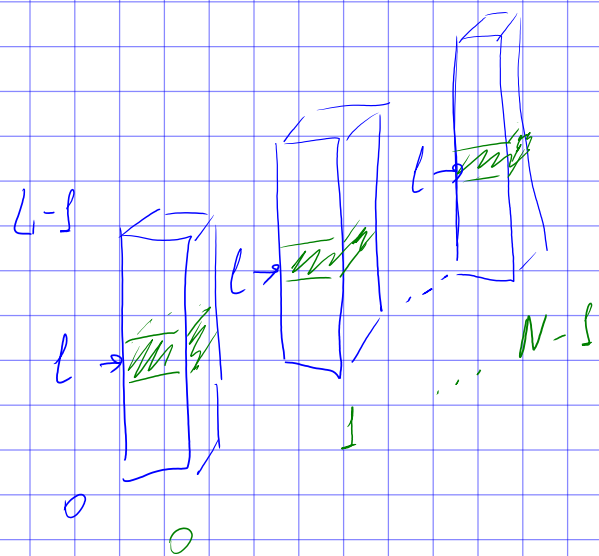
One possible way is to add "layer fusing" or
 "layer communication" step before regular
 RNN cell dynamics:



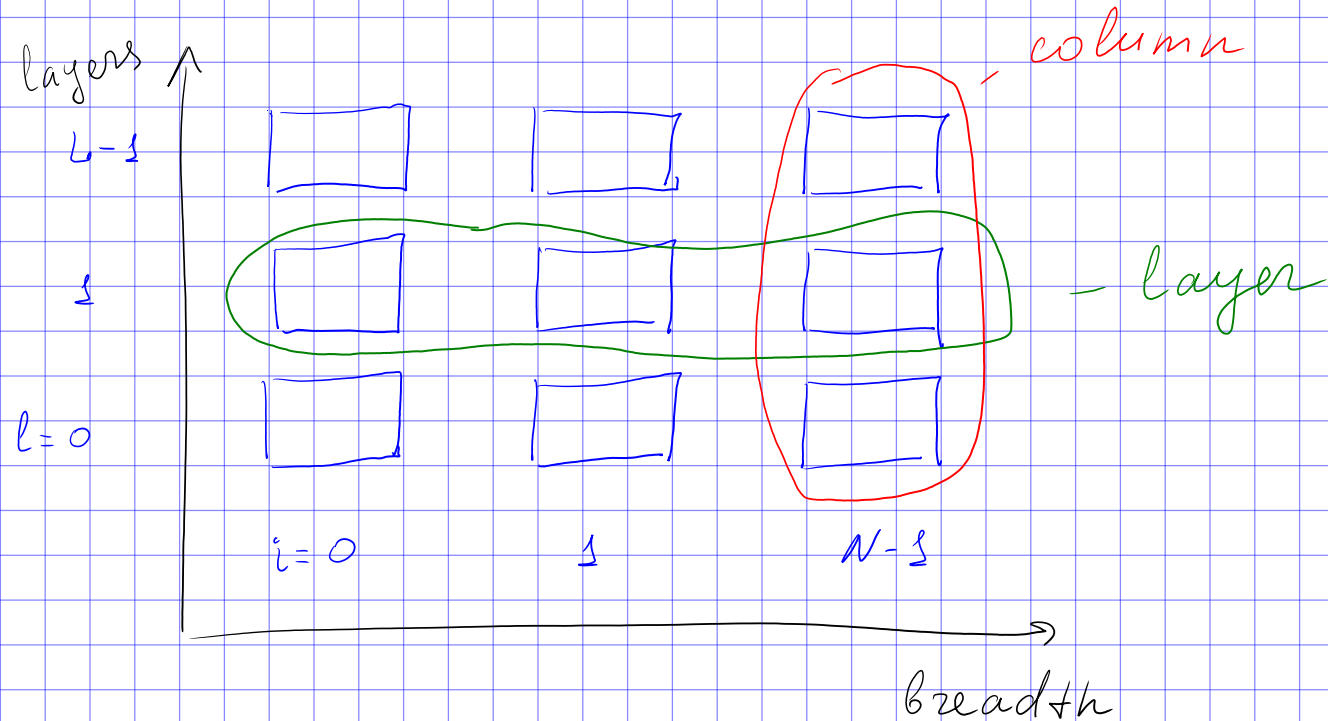
where  is:

$$\tilde{h}_{t-1}^{l,i} = \text{Attn}(h_{t-1}^{l,\bullet}) = \text{Attn}(h_{t-1}^{l,j} \text{ for } j \in [0, N])$$

i.e. Attn connects hidden states within the layer l (over breadth axis)



⑥* Let's check in more details full hidden state H_t :



I imagine this model
in the RL context

$\Rightarrow$ it somehow should
accept observations o_t
and select actions a_t



let's talk about
columns next

6.1* Until now each column I/O was not defined

There are several options:

a) input can be external or internal, i.e. our model may observe both an environment and the model's self output

the same is with output: external - to env (action)
internal - to model

⇒ inter-column communication (besides attention)
may be organised via self-observation
 i -th column $\rightarrow$ j -th column
output input

6.2

b) Output of each column may or may not be "clamped" explicitly and directly with some loss, e.g.:

- policy/value heads on top of the column output with corresponding Policy Gradient / DQN losses
- forward model "next obs prediction" with reconstruction loss
- etc.

While some columns are entirely "intermediate" without a loss on their output. Or even w/o an output!

6.3

As a result, we may treat columns as "bound" / "free" in terms of their role specialisation, which depends on whether their input/output has pre-specified fixed meaning/role.

We dictate the model what we want from bound columns and let model decide how to offload the computations/storage to free columns.

⑦ Some additional notes:

a) I assume model to be inherently multimodal: observation parts (each part passed to a separate column) can be of diff modalities, columns deal with/work with/specialise on non-homogeneous information

⇒ same is for columns output including actions; we may support multipart actions (motor, text, vis)

⇒ that's why cross-attention may be suitable too

7.1 The perfectly columnar structure with the specified form of connections (intra-columnar hierarchy, within-layer cross-columnar attention and cross-columnar output-to-input) is a special form of more general topology (like small-world interconnected Echo State Machine on the opposite extreme of the topology spectrum)

I selected more regular option with more pronounced columnar factorisation as a simpler initial choice. which may be enough to test hypotheses